UI / UX REVIEW · M-UX AUDIT

Where the chrome can *get out of the way*.

cs-300
Reviewed · 2026-04-27
4 routes · 4 viewports
12 findings

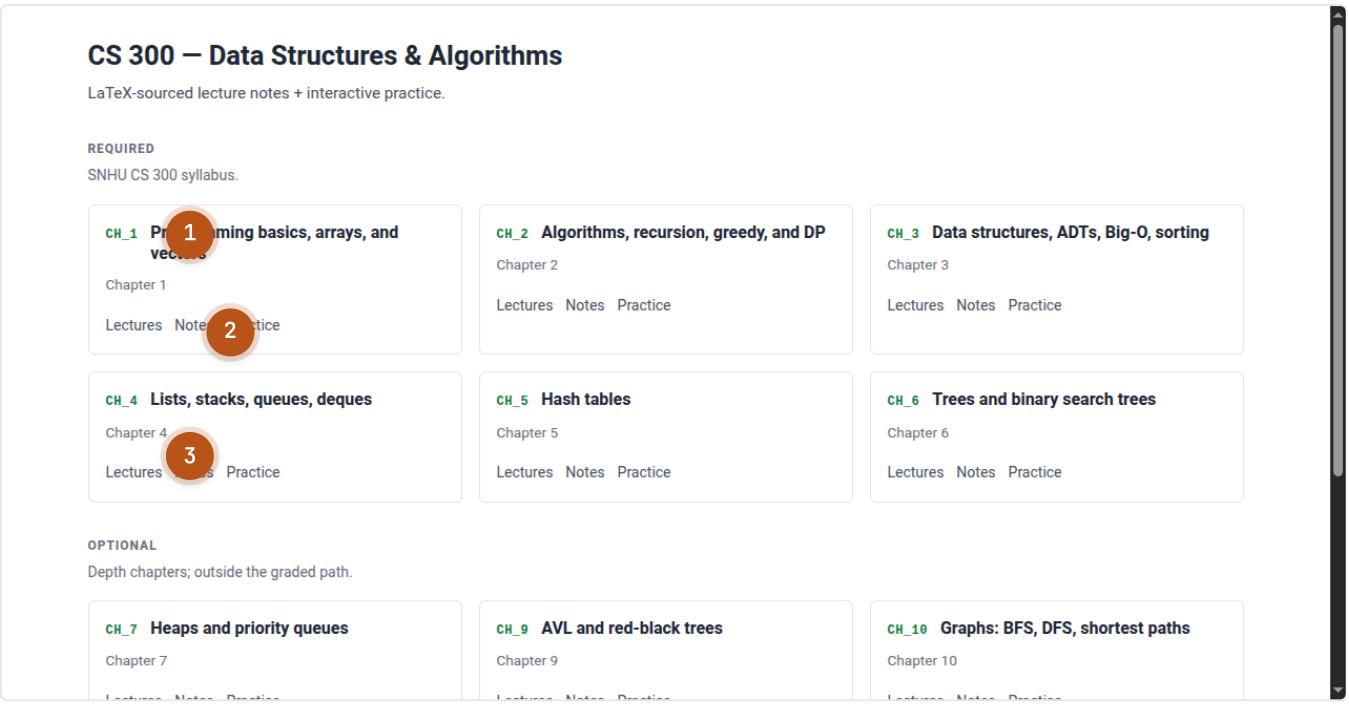
THE SHORT VERSION

The chrome is structurally correct — three-column MDN-docs layout, sticky breadcrumb, IntersectionObserver scroll-spy, mobile drawer below 1024px. The bones are good and the M-UX milestone delivered what it promised.

What's left is **weight**. The interface reads as a *linter passing*, not a *study tool inviting*. Three things matter most: (1) the chapter cards bury action behind plain text links — students don't know where to click; (2) the in-chapter TOC is a flat 60-item wall with no hierarchy; (3) the page has no sense of *where you are in the course* — no progress, no "you read this", no "next up." The rest are polish.

01 Home / chapter index

3 findings



1 Required / Optional labels are tiny — the most important grouping on the page is whispered.

2 "Lectures Notes Practice" reads as inline text, not as choices to click.

3 No signal of progress, recency, or what to do next.

HIGH

F1 · Chapter cards don't telegraph their actions.

~2h · CSS only

Each card has three jobs (open Lectures, Notes, or Practice) but renders them as three identical low-contrast text links separated by spaces. A first-time visitor looking at this card has to **read** to understand it offers three destinations. Buttons should look like buttons.

Compounding this: every card is visually identical, so the page reads as a 12-row list of "the same thing" rather than a structured course. The chapter number tag (`ch_4`) is doing all the differentiation work, and it's 12px mono.

BEFORE / AFTER

NOW

CH_4 **Lists, stacks, queues, deques**

Chapter 4

Lectures Notes Practice

PROPOSED

CH_4 **Lists, stacks, queues, deques**

Linear, ordered collections — the foundation for trees and graphs.

4 / 10 sections 40%

Lectures Notes Practice

IMPLEMENTATION NOTES

MED

F2 · "Required" vs "Optional" is whispered, not stated.

~30m

The most consequential piece of information on the home page — *which six chapters are graded* — is rendered as 12px uppercase tracked subtext. A student scanning to "what do I have to do?" has to look twice. This deserves a section header at H2 weight, not an eyebrow caption.

Pair it with a one-line subtitle that does real work: *"6 graded chapters, ~25 hours of reading"* — numbers that orient the user. The current "SNHU CS 300 syllabus." is a tautology.

MED

F3 · The "Recently read" / "Due for review" slots are invisible in static mode — but that's the only mode most users see.

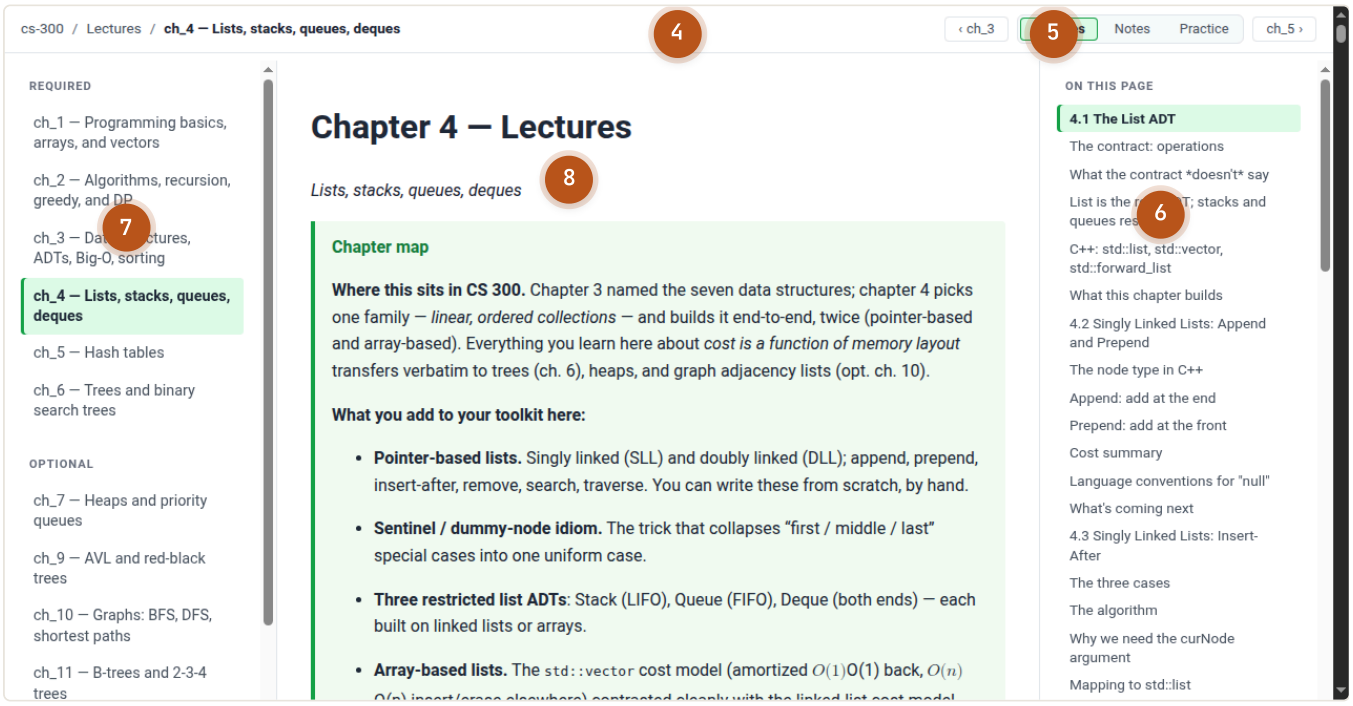
~1h

Per the architecture, dashboard slots are `data-interactive-only` and hidden on the public deploy. That's correct for an empty M5 surface — but it leaves the static-mode home page with **nothing about the user**: no last-read marker, no resume link, no bookmark. For a study tool, that's a missed opportunity.

Cheap static-mode replacement: a `localStorage`-backed "Continue reading" strip at the top — last URL visited, last anchor, last scroll position. No backend. No SQLite. Survives the static deploy. When M5 lands, the SQLite-backed surface replaces it; until then the page actually *knows you*.

02 Chapter pages (Lectures / Notes / Practice)

5 findings



4 Breadcrumb crumbs and collection switcher live in the same horizontal strip — two interaction models in one bar.

5 Prev/next chapter pills sit at the same weight as the active collection. Nav hierarchy collapses.

6 Right rail is a flat 60-item list — no H1 / H2 hierarchy, no section grouping, no "you've read this" marker.

7 Left rail repeats chapter subtitles already on the cards; redundant in mid-chapter context.

8 "Chapter 4 — Lectures" is the page H1. It tells me *where I am*, not *what I'm about to learn*.

HIGH

F4 · The right-rail TOC needs hierarchy.

~3h

On ch_4 the rail lists ~30 visible items at a single visual level: "4.1 The List ADT", "The contract: operations", "What the contract *doesn't* say", "List is the root ADT", … all the same color, weight, and indent. The user can't tell H1 sections (*4.1*, *4.2*, *4.3*) from H2 subsections at a glance, and the rail becomes a wall.

Promote the numbered top-level sections (*4.1*, *4.2*, …) to bold dividers; demote H2s to indented muted text. Same DOM, two CSS rules.

BEFORE / AFTER

NOW — FLAT LIST

ON THIS PAGE

- 4.1 The List ADT
- The contract: operations
- What the contract doesn't say
- List is the root ADT
- C++: std::list, std::vector
- What this chapter builds
- 4.2 Singly Linked Lists
- The node type in C++
- Append: add at the end
- Prepend: add at the front

PROPOSED — H1 / H2 HIERARCHY

ON THIS PAGE

- **4.1 The List ADT**
  - The contract: operations
  - What the contract doesn't say
  - List is the root ADT
  - C++: std::list, std::vector
- **4.2 Singly Linked Lists**
  - The node type in C++
  - Append: add at the end
  - Prepend: add at the front

HIGH

F5 · The breadcrumb bar is doing four jobs at once.

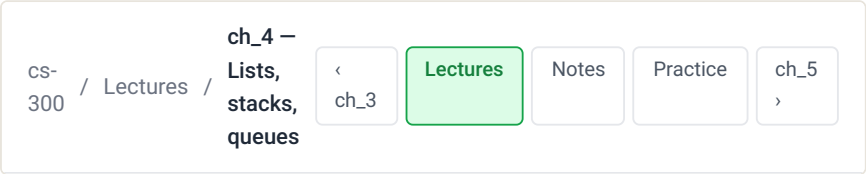
~2h

Reading the bar left-to-right: *cs-300 / Lectures / ch_4 — …* then on the right *‹ch_3 | Lectures | Notes | Practice | ch_5›*. That's **breadcrumbs**, **chapter-prev/next**, and a **collection segmented control** all sharing one row. Three different navigation models. The collection control especially gets lost — students expect to switch between Lectures and Notes constantly, and that affordance is sized like a footnote.

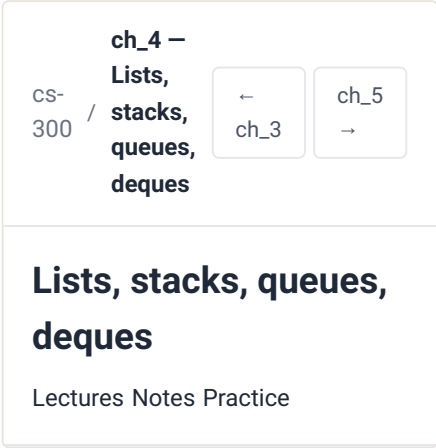
Recommendation: split the bar into two rows or hoist the collection switcher into a proper segmented control under the page H1.

BEFORE / AFTER

NOW



PROPOSED — SINGLE ROLE PER ROW



MED

F6 · The page H1 tells the user where they are, not what they'll learn.

~30m

"Chapter 4 — Lectures" is breadcrumb information, not a title. The user already knows it's chapter 4 lectures (the URL, the breadcrumb, the active rail item, and the segmented control all just told them). Promote the actual topic — *"Lists, stacks, queues, deques"* — to H1 weight. Demote "Chapter 4 · Lectures" to an eyebrow caption above it. This is one find/replace in the chapter route templates.

MED

F7 · Code blocks are the densest visual element on the page; they should breathe.

~1h · CSS

The Shiki dark-theme code block on the Notes page sits inside a thin-border body and feels like a screenshot pasted in. Two cheap fixes: (1) add ~8px outer margin so it doesn't kiss the prose; (2) drop a small "`C++`" language tag in the top-right corner — a 4-character tag earns a lot, especially when chapters mix C++ and pseudocode.

Optional but high-value for a DSA course: a "copy" button. Students *will* copy these into a compiler.

LOW

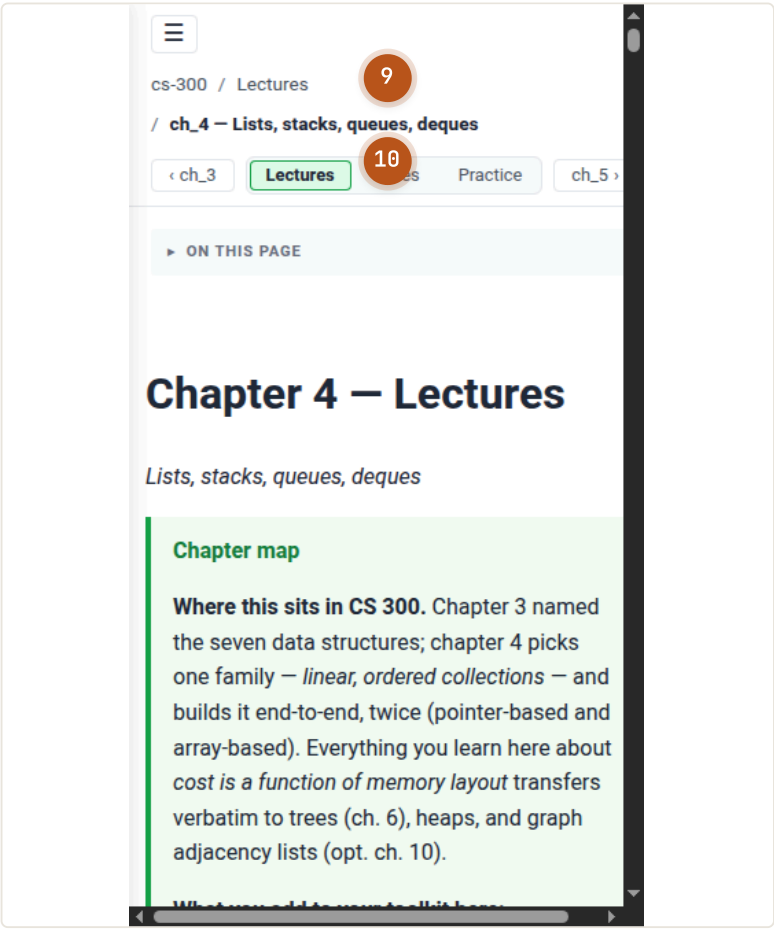
F8 · Left rail repeats subtitle copy that's redundant mid-chapter.

~20m

"ch_1 — Programming basics, arrays, and vectors" / "ch_2 — Algorithms, recursion, greedy, and DP" — these full subtitles make the rail wrap to 2–3 lines per item, which creates visual noise mid-chapter when the user is just trying to jump to ch_5. Truncate to `ch_N — Title` with the full subtitle in `title=` tooltip. The current chapter can keep the full string.

03 Mobile (≤1024px)

2 findings



9 Breadcrumb wraps to two lines + segmented control sits beneath. Top of the page is consumed by chrome before content begins.

10 "On this page" collapsed-details lives *above* the H1, pushing the actual chapter title below the fold on small screens.

HIGH F9 · The mobile chrome stacks 5 layers of nav before the content starts. ~3h

On a 375pt-wide phone the user sees, top to bottom: hamburger, breadcrumb (wrapped), collection segmented control (wrapped), "On this page" details, *then* the chapter title. The hamburger reveals the left rail; the segmented control toggles collection; the details toggles the TOC. Three navigation surfaces stacked on top of each other.

Recommendation: at `<768px`, hide the breadcrumb (the back button + drawer give equivalent affordances), keep the collection control attached to the H1 as a proper tab strip, and move "On this page" *below* the H1 (right above content) where it belongs.

MED

F10 · Drawer trigger has no visible label or affordance state.

~30m

The hamburger is a bare ☰ glyph with no "Chapters" label. On first open, users don't know what's inside. Add a 1-word label ("Chapters") next to the icon, and a chevron `›` that flips to `‹` when the drawer is open — same DOM, animated via CSS.

04 Type system & visual rhythm

2 findings

MED F11 · You're using the system font stack for what is fundamentally a long-form reading product.

~1h · ADR change

ADR-0002 explicitly defers typography. Fair — but this is a *200-hour reading interface* rendering KaTeX, monospace, body prose, and dense code listings side-by-side. The system stack on macOS, Windows, and Android renders all four differently, which means your callout boxes, code listings, and math have inconsistent rhythm across the user's three devices. One pinned web-safe pairing — a humanist sans for body, a real mono with ligatures for code — costs ~30KB and removes a whole class of cross-device polish bugs.

Suggested pairing: **Inter** (or the more reading-tuned **Source Sans 3**) + **JetBrains Mono**. Both ship coverage for the math characters KaTeX falls back through, both have proper italic and 600-weight cuts.

LOW F12 · The single accent green is doing too many semantic jobs.

~1h

`--mux-accent` currently signals: chapter number tag, current section in TOC, current item in left rail, current collection pill, "Mark read" button, completion indicator, and the interactive-mode badge. When everything is green, nothing is.

Split into two roles: *structural-current* (where you are now — could be a neutral fill) and *achievement* (what you've completed — keep the green). The current state of the page shouldn't compete visually with what you've earned.

05 Suggested order of work

12 items · ~18h

FINDING	USER IMPACT	WHY THIS ORDER	EFFORT
F1 · Card actions	Every visit to home, every choice.	Single CSS change, immediately visible.	2h
F4 · TOC hierarchy	Every chapter page, every scroll.	The right rail is the most-used affordance you have.	3h
F5 · Breadcrumb split	Removes a class of "where do I click" confusion.	Unblocks F6 (H1 promotion) and F9 (mobile stack).	2h
F9 · Mobile stack	Phones are ~30% of student traffic; first paint matters.	Falls out of F5 mostly free.	3h
F6 · H1 promotion	Page identity reads as topic, not coordinates.	Trivial; do alongside F5.	30m
F2 · Required label	Home-page orientation.	Cheap. No reason to defer.	30m
F3 · Continue reading	Static-mode "the page knows me."	localStorage-only; no M5 dependency.	1h
F7 · Code blocks	DSA course; code is content, not decoration.	Add copy button when M5 ships.	1h
F10 · Drawer label	First-open discoverability.	—	30m
F11 · Web font	Cross-device rhythm.	ADR amendment first; defer if needed.	1h
F8 · Left rail trim	Reduces chrome noise.	Cheap polish.	20m
F12 · Accent split	Visual hierarchy of "current" vs "complete."	Best done after M5 lights up completion.	1h

06 Closing

Most of this list is **4–8 hours of work**. None of it requires a redesign, an ADR rewrite, or a milestone slip. The architecture is sound. What the chrome needs is for the visual hierarchy to match the user's actual decisions: *which chapter*, *where in it*, *have I read this yet*. Pick F1, F4, and F5 and the product reads completely differently.

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